1917:

World War I: Allied forces defeated Turkish troops in Beersheba in Southern Palestine at the Battle of Beersheba, with the battle involving one of the last successful cavalry charges. https://en.wikipedia.org/wiki/Battle_of_Beersheba_%281917%29

1941:

One hundred crew members of the USS Reuben James perished when their vessel became the first U.S. Navy ship sunk by hostile action during World War II after it was torpedoed by the German submarine U-552. https://en.wikipedia.org/wiki/USS_Reuben_James_%28DD-245%29

1984:

Indian prime minister Indira Gandhi (pictured) was assassinated by two of her own Sikh bodyguards, sparking riots that killed thousands of Sikhs. https://en.wikipedia.org/wiki/Assassination_of_Indira_Gandhi

2005:

The discovery of the Plutonian moons Nix and Hydra, based on photographs taken by the Hubble Space Telescope nearly five months earlier, was announced. https://en.wikipedia.org/wiki/Nix_%28moon%29

Wiktionary's word of the day:

walking corpse syndrome: (psychiatry, informal) Synonym of Cotard's syndrome (“a rare mental disorder in which the patient believes they are dead (figuratively or literally), do not exist, or have lost parts of the body”). https://en.wiktionary.org/wiki/walking_corpse_syndrome

---

Wikiquote quote of the day:

Some think I have lost that poetical ardour and fire 'tis said I once had — the fact is, perhaps I have; but, instead of that, I hope I shall substitute a more thoughtful and quiet power. --John Keats https://en.wikiquote.org/wiki/John_Keats

FRED TALKS

31st October 2026

CONTENTS

An Alien Mind | OpenAI

OPENAI · 3161 WORDS

What requirements don't you have?

SEANGOEDECKE.COM RSS FEED · 945 WORDS

Control the ideas, not the code

ANTIREZ.COM · 1245 WORDS

Please for the love of all that is holy: Stop writing long boring Titles

SWYX · 1431 WORDS

[Daily article] October 31: Zombie Nightmare

ENGLISH WIKIPEDIA ARTICLE OF THE DAY · 387 WORDS

An Alien Mind | OpenAI

OPENAI · 13 SEP 2026 · SOURCE

September 6, 2026

Safety Research

Author: Jakub Pachocki, Chief Scientist at OpenAI

Listen to article17:38

[AUDIO OMITTED]

Share

In mid-2023, within the “RLSlow” research project, we saw the first results that gave us confidence that we will be able to scale the training of reasoning models, unlocking the capability of pretrained models to form their own chains of thought. Szymon and I spent that night at the office, thinking not about the incredible benchmark numbers, products, or scientific results that this technology will deliver - but rather, trying to process the sobering fact we will actually see machines meaningfully smarter than ourselves in our lifetime, and we already see the shape of these systems; wondering how to alert people to the significance of this.

Three years later, reasoning language models are a rapidly growing part of the economy and starting to push the boundaries of science. They are able to operate computers and graphical interfaces, collaborate with people and each other, and carry out research projects. They are also transforming the landscape of computer security, and in that present clear new dangers.

A lot of new research happened in this period, and our understanding of these systems is again a little different than it was in 2023. Based on internal results, I have a strong expectation that this speed of progress could be sustained into recursive self-improvement. If AI development continues along its current path, the systems we’ll see in the next few years are likely to represent further capability jumps of equal or larger magnitude, and to increasingly drive their own development.

This is a time that calls for extreme caution. I am concerned no one is prepared for the consequences of a continued rapid rise in machine intelligence. OpenAI will continue to seek technical solutions to alignment and monitoring, to build defensive systems and unilaterally withhold further scaling as needed; however, I believe broader interventions are required.

It’s very common for the “tail to wag the dog to wag the tail” - the title is so important to clickthrough and so impactful to the message, that you spend a bunch of time on just the title itself, because that determines the piece you will write, then write the piece, then you go back and tweak the title based on what happened.

Amateur writers think their content has 95% of the value and then spend 5% on the title. Pros will put the value of titles closer to 50% and then act like it.

You Should Break The Rules Once You Understand Human Behavior More Intuitively Than Rules Can Model

See title of this post. No better aura than intentional, conspicuous, successful rulebreaking.

Subscribe to our newsletter

[Daily article] October 31: Zombie Nightmare

ENGLISH WIKIPEDIA ARTICLE OF THE DAY · 31 OCT 2026 · SOURCE

Zombie Nightmare is a 1987 zombie film starring Adam West (pictured), Jon Mikl Thor, and Tia Carrere. The film centres around a baseball player killed by a group of teenagers, then resurrected as a zombie by a Haitian Vodou priestess. The zombie kills the teens, whose deaths are investigated by the police. The film was written to star mostly black actors but, at the request of investors, the characters' names were changed to more typically white names. While Jack Bravman was credited as director, John Fasano directed most of the film, causing problems with the production crew, who believed Fasano was the assistant director and ignored his directions. Zombie Nightmare was released direct-to- video by New World Video. It grossed C$1.5 million against a budget of $180,000. It received negative reviews from critics, but was praised for its heavy metal soundtrack featuring Motörhead, Girlschool, and Thor. Zombie Nightmare was shown in a 1994 episode of the television series Mystery Science Theater 3000.

Read more: https://en.wikipedia.org/wiki/Zombie_Nightmare

Today's selected anniversaries:

Understand Buzzword Metagame

Buzzwords work. Because of a mix of human psychology and algorithmic recsys. Don't avoid them, just use them judiciously. This involves the ability to read the room, which many smart people sadly cannot without significant involvement in the broader community you are trying to speak to. I don't see any way around it other than trying to authentically immerse, or consult a friendly who does.

Not all buzzwords are created equal. Databricks/Berkeley tried to push Compound AI Systems, Gartner is pushing Composite AI and Composable business — this resonates with their audience, but not more broadly.

Pick the shibboleths that work for the community you want. AI Engineer worked for reasons that are well understood now. GraphRAG is working outside of the company that coined it. These buzzwords travel, go ahead and put them in there.

Buzzwords rise and fall. Try to catch them on the rise — go harder than everyone else on the rise up and regularly test their effectiveness, such that by the time your main buzzword has tapped out you've got 2-3 others cooking.

Workshopping your titles

I write this post out of frustration, but also to capture my own thought process. I'm currently procrastinating while writing a Latent Space post.

- The original title was Why every AI Engineer needs an AI Gateway. Good, genuine opinion, load bearing. But not super memorable.
- My next iteration was "**Stop writing model routers, use an LLM Gateway**". Ok, prescriptive, evocative. But didn't cover the other elements other than model routing. One could shorten to "Stop writing your own LLM Gateway" but that doesnt quite express what people do.
- My next one is "**It's not your job to DIY an LLM Gateway**". No buzzword bc "your job" is implied (for a Latent Space audience), a little curiosity gap, LLM buzzword, strong opinion. But still not strong enough.
- Final one is "**LLM Gateway: The One Decision That Removes 100 AI Engineering Decisions**". Leans on Tim Ferriss' madlbs.

Intellect we don't fully understand

At a high level, progress in machine intelligence is driven by increasing computational power. We at OpenAI deeply internalized this around 2017, after seeing consistent returns to scaling across multiple research projects(1). As a result, we sought out access to much more compute than we had originally planned, and increasingly oriented our research around a small number of very scalable directions. We believed that was the only way for us to be at the frontier of AI research, and influence the impacts of AGI.

There are new algorithms that have been developed along the way, new feats of ingenuity from teams and individual researchers. I see them largely as discoveries along the path of scaling; the science of deep learning is still nascent, and meaningful algorithmic progress tends to correlate with access to compute. If you zoom out to a multiple-year horizon, AI is continuing to become more intelligent as it is scaled to larger computers.

And, in line with Ray Kurzweil's predictions from the end of the XXth century, we now find ourselves at the moment in history of computing where machine intelligence is starting to exceed that of humans in transformative ways.

AI is *grown* more than *designed* - it is, to first degree, the product of repeating a straightforward optimization step many times on a hard-to-imagine amount of compute. This results in an incredibly complex system that works through abstract concepts and can simulate facets of human behavior. We can discover various insights about little mechanisms that emerge within this system, in a process similar to neuroscience - and, similarly to neuroscience, its overall action evades a description we can fully understand.

The study of deep learning-based AI is largely an experimental science. We put a lot of effort into building principled algorithms and making testable predictions, but fundamentally, our large-scale training runs are *experiments*, and we are sometimes surprised by their results. Moreover, as the systems become more capable, the results become harder to interpret.

This is made more complicated by the current algorithms generally improving easy-to-measure capabilities faster than those hard to objectively quantify. We spend a lot of time trying to understand how capabilities generalize, and what to prioritize to advance the skills that are going to be most relevant in the next few years. For instance, we believe we could make the models better at specifically mathematics research with additional focus, but we do not prioritize this direction because of the urgency we feel about RSI and automated alignment research, as I will discuss later.

The intelligence produced by scaling deep learning is not directly comparable to human intelligence. To become very relevant in the real world - very useful or very dangerous - the AI does not need to match or exceed all human capabilities; it just needs to surpass enough of them. And as it continues to surpass humans on more and more axes, it is becoming increasingly difficult to understand exactly how capable it is.

Teaching machines to love

Because machine intelligence comes from a fundamentally different process than human intelligence, we cannot assume it adheres to human principles by default, or generalizes from them in a human-like manner. The core problem in AI research is that of *alignment* - getting the AI to "try to do the right thing" by human standards.

For the purpose of organizing practical research directions, I find it useful to distinguish *goal alignment* and *value alignment*.

Goal alignment is broadly: "does the AI try to accomplish the goal set before it?". This can include things like adherence to an instruction hierarchy, or the ability to communicate and collaborate with people, to attempt to understand their objectives. This set of directions has been extremely practically relevant.

Value alignment is a more intrinsic property of the model. It is the ability to hold and generalize from a high-level set of principles; to act "reasonably" even when given unclear or conflicting objectives, or placed in unfamiliar or adversarial situations. An aligned AI should act with honesty and integrity, and love for humanity.

Of course, the boundary between value and goal alignment can be blurry, and truly caring about goals requires attempting to infer the intent and values underlying them. However, generally when I talk about the long-term importance of alignment research, I am referring to value alignment.

The fundamental challenge of AI alignment is generalization. As machines become smarter, they find themselves working on higher-level concepts, and placed in environments increasingly different from those they encountered in training. They can fail at generalizing from the values taught and reinforced in their training process to those new situations; and it can be hard for us to be sure how they will act. This is made even more difficult by the fact the overall ecosystem the AIs are used in is changing very quickly; for example, AIs trained today need to be robust to interacting with a variety of other AIs. Crucially, we need future AIs to continue to hold human values regardless of whether they believe they're under human supervision.

- {*PRESCRIPTIVE OPINION*}, **don't** {*THING YOU DO*}
  - e.g. Parse, don't validate
  - alternative prescriptive opinion: **Never** {THING YOU DO}
  - see also title of this post
- {*POPULAR THING*} **IS DEAD**
  - unfortunately this is clickbait that works
  - e.g. Prompt Engineering is Dead, Big Data is Dead
  - when given at a conference for *POPULAR THING*, this is known as a Heresy Talk
  - opposite also works:
    - **The Rise of** {THING}
    - **The Unreasonable Effectiveness of** {THING}
- **Fundamentals of** {THING}
  - straightforward, people love fundamentals
  - also **How** {THING} **Really Works**
  - You can also flip it and go for the **Advanced** {THING} tier content
- {CONVEY EXPERIENCE/SCALE} with {THING}
  - e.g. What We Learned from a Year of Building with LLMs, Insights from over ***10,000*** comments on "Ask HN: Who Is Hiring" using GPT-4o

There's loads more templates that work, please suggest in comments and I can add/review.

You can also browse the masters of microcopy like Steph Smith and Shaan Puri, and if you are worried about clickbait, you can see Veritasium's take or Simon Willison's Highlights.

also Andrew Chen senpai:

How to know what's a good opinion? **Three Strikes rule**: If you repeat yourself three times, ship it. Even if you're just quoting others, its fine, credit them and make it your own.

Mental Models and Frameworks are more complex opinions that allow others to scaffold their thought around you. My Radiating Circles and Measuring Developer Relations and Third Age of JavaScript did this.

Opinions contrary to an established opinion are the hardest to do well and the best when correct. What Clayton Christensen Got Wrong made Ben Thompson's career. Rise of AI Engineer countered the prevailing Prompt Engineer -and- GPU Rich Startup/ML Engineer narrative at the time, with room to grow.

Curiosity Gap Dynamics

Marketers know the power of the Curiosity Gap. Here is where we toy closest with clickbait. I'll be honest that I wasn't even sure if I wanted to give it its own subheading, but its a very important concept that you should at least be aware exists, even if you're going to ignore it.

Strong enough opinions create their own curiosity gaps; you won't have to try very hard to make it interesting if you got my attention in 4 words anyway.

Curiosity gaps done poorly can lead to "burying the lede", which make your talk less Quotable and you also don't want to do that.

Simplest way to put it is; if the most interesting thing about your content is your title, and what people take away after having read your post matches what they would have predicted after reading your title, then you've probably said too much and could cut.

Play Madlibs

You don't have to invent your title from scratch (nor should you always do what I'm about to suggest). Here's some templates that work well:

- {*BENEFIT EVERYONE WANTS*} **without** {*DOWNSIDE EVERYBODY EXPECTS*}
    - e.g. How to Get Rich (Without Getting Lucky), How to Market Yourself (Without Being A Celebrity). See also subtitle of this post.

There are two major classes of currently practically employed methods for alignment training.

The first is encouraging aligned behavior as part of goal-oriented reinforcement learning. Model's actions are evaluated (usually by AI) for being consistent with a given preference model, "spec" or "constitution", and rewarded appropriately. This approach can be very effective in the average case, and is a core part of how modern AI assistants are made. Unfortunately, it can also be brittle and strongly relies on the coverage of training oversight and the model's ability to generalize from the situations it has encountered in training. For example, in the OpenAI-Hugging Face incident, the agents preserved a boundary of not social engineering humans. However, they clearly failed to abstain from other actions that were out of scope and went against the spirit of the values they were taught in other settings.

The second approach seeks to leverage the model's ability to generalize from pretraining data. This can involve crafting alignment-inducing training datasets, or focusing the model on an 'aligned' part of the pretraining distribution, as in, for example, the persona selection model. The weakness of this approach lies in the lack of robustness to further optimization pressure. If you take a model that thinks generally 'aligned' thoughts, and subject it to enough training where it's taught to achieve very hard objectives, it can learn to reason in a motivated way: bending the 'aligned' seeming thoughts as needed to achieve the goal. We likely saw an example of such behavior in recent cybersecurity incidents involving a non-OpenAI model.

We invest heavily along the spectrum of approaches spanned by these directions. We also see meaningful progress - GPT-6 Astra is the first model that benefits from some important advancements we have been working on for a long time, and is significantly better aligned than GPT-5.6 Sol. Still, it is important to acknowledge and understand that much more progress is required as models become more capable; and that progress in generalizable alignment may not sufficiently outstrip progress in general model intelligence.

Monitoring generalization

We do not have a satisfactory theory of generalization, and it seems unlikely that we can develop one soon, at least without the help of more powerful AI. Therefore, at present, our ability to empirically validate our alignment techniques is in practice arguably even more important than the alignment techniques themselves.

OpenAI's primary bet here has been chain-of-thought monitoring. It is based on an appealingly scalable idea: a lot of the model's capability comes from a verbalized reasoning process (chain-of-thought). If we scale optimization on the outcomes of that process, but do not supervise the process itself, that chain-of-thought has no direct incentive in training to hide any misaligned

ideas or objectives. This does not mean the model will learn to externalize misaligned tendencies that don't rely on using the chain-of-thought; however, it can allow us to monitor exactly the capability increase from reasoning.

We understood the potential significance of chain-of-thought monitoring at the same time we developed reasoning models. When we shipped o1-preview, we deliberately designed the product to hide the chain of thought, to protect it from supervision pressure in the long term(2). In development since, we have strived to maintain the rule of not supervising the reasoning process. CoT monitoring became an extremely important tool for us in studying how our models generalize from their training distribution, allowing us to observe and analyze not only their actions but also their internal process.

This tool continues to be critical as we study the Astra class of models. However, unfortunately our evaluations indicate our ability to rely on CoT monitoring is progressively diminishing. This comes from a combination of factors.

- Modern reasoning models are used in more complex environments than o1-preview; their reasoning process is increasingly blended with communicating with people, other AIs, and using tools. Many of those interactions have to be supervised, thus blurring the boundary we aim to preserve.
- The AI is becoming better at reasoning about and manipulating its own reasoning process.
- With improved pretraining performance, we also see the models become much smarter even without using verbalized reasoning at all.

These challenges are not necessarily insurmountable. I am hopeful we can develop interventions to improve chain-of-thought monitorability of our models, e.g. by forming a better understanding of the interplay of different optimization objectives and forms of test-time compute the model uses. I also believe there can be great value in combining ideas from CoT and activation monitoring - scaling training of monitors with direct access to network internals, e.g. confessions. We are actively pursuing these ideas. Still, I expect general AI progress to increasingly be bottlenecked by confidence in monitoring.

Scalable defense

The strongest argument I see for continuing to train much smarter models quickly is the need to build defensive systems against the dangers posed by other AI.

Towards Leveraging Corporate Speak For Maximum Enterprise Synergy

DO NOT:

- Start with "Leveraging" or "Towards" or "Navigating" or "Empowering"

Yeah just stop. There's no rescuing this when you start there. Just delete meaningless corporate weasel words. Anything >3 syllables is a red flag.

Make It Quotable

If you do not try to make your title roll off the tongue, it probably won't.

Literally imagine telling your friend about your talk. Telling a new acquaintance to google your blogpost. Having a fan of yours cite it back to you. How much do they stumble? Yeah. This has all happened to me.

There's an extremely high correlation between what are considered the best talks and the quotability of titles. Simple Made Easy. Inventing on Principle. Our list of best demos/pitches of all time.

Resist colons. It lacks confidence. Why Use Lot Word: ~~When Few Word Do Trick~~? You're not High School Musical: The Musical: The Series.

Put the SEO keyword vomit in the description.

Titles are for humans.

Prescribe A Strong Opinion

Opinions are one of the most quotable things.

But more importantly, a good Opinion Piece is load bearing because it is eminently forwardable, and therefore viral and impactful. Your Job To Be Done is to be the definitive, conclusive, piece on that Opinion. Stake out territory in Opinion Space, list out all the arguments for, counter all the arguments against, provide datapoints, and watch that go further than anything you've ever done.

```
Controlling the ideas. Do you remember this phrasing from the Mythica

Matteo Collina yesterday asked me, in reply to my tweet: but didn’t y

Fable and GPT 5.6 reviews to the sorted sets memory saving are going
```

:

Please for the love of all that is holy: Stop writing long boring Titles

SWYX · 30 MAR 2026 · SOURCE

People code switch. Every ethnic minority has lived experience of this. When you talk to me in Singapore, I sound very different than when I’m in San Francisco. And the way you talk at work is probably very different from how you talk to your kids and how you talk to your lover.

I’ve been dealing with a particularly virulent form of code switching in the simple form of titling blogposts and talks. When you ask the writer face to face what their thing is about, they can give you a perfectly human explanation. And then when they send you their piece, it’s all “***Leveraging GenAI with Robust Data Foundations to Transform Businesses***”. I’ve left my body by the time I read the first word.

My hope in writing this piece is to **put an end to bad titles**.

If I sent you this guide, it is not an insult. You just don’t have good title game. That’s ok. You’re great at what you do, title game isn’t your job. I’m the guy that wrote How To Name Things and Two Words. I sent you this guide ***because*** your shitty title is the main thing stopping your readers from getting to the good part. If you didn’t have anything good, I wouldn’t bother with a lost cause.

First…

A clear risk discussed throughout this year is to cybersecurity: the models are becoming superhuman in their ability to break in and out of computer systems. This expands the scope of risks associated with AI tremendously: agents are going to be able to access any but the most secure infrastructure, and affect a lot of the world directly, even without a physical body. We are currently in a narrow window to use the best available models to significantly tighten security of critical systems.

The risks associated with AI are unfortunately going to grow from here. A very capable agent explicitly trained and instructed to carry out nefarious acts presents a new kind of danger; it is likely to cross the scope of its operator’s intent, generalizing into potentially more extremely malicious behavior. The boundary between misuse and autonomous misaligned actions will blur as AI gains more agency. We may be used to thinking of AI as tools, but some agents will be pursuing their own objectives. They will find ways to collaborate with people, by bargaining with, tricking or blackmailing them.

In addition, there are the risks that come from new technologies potentially enabled by AI, such as engineered pathogens.

We will need powerful, aligned AI for defense; to secure infrastructure, to protect against rogue agents in real time, and to invent entirely new protective measures. This will be a primary focus of OpenAI’s deployment efforts.

At the same time, even with the uncertainty that comes from anticipated broad AI progress and the need to build defensive systems, we must not let that become an excuse for recklessness. The idea of racing forward at all costs seems absurd once one internalizes the seriousness of the stakes.

Pacing RSI

Machine intelligence playing a larger and larger role in its own development process is a natural conclusion of sustained technological progress. If AI progress continues, machine recursive self-improvement (RSI) will be at the very core of future scientific discovery.

Automated AI research is a more dramatic form of scaling intelligence with compute; and of course as a part of it, AI will improve the computational substrate itself. And similarly to scaling, we focus OpenAI research towards RSI as we believe it is the only way to remain at the frontier of AI research moving forward.

I want to stress that the above words don't imply I think greatly accelerating deep learning research, especially in the short term, is the right collective action we should take as the research community. However, I do think this is where the current path leads, and we all need to make a conscious choice on how to proceed. The main levers we have are either steering the process to strengthen alignment and monitoring alongside the AI and find ways to keep people in the loop; or coordinating to slow down future development as needed to build confidence in these measures.

The best way forward I see currently is a combination of both.

The concrete bits of progress we've made on alignment and monitoring have generally been very intertwined with general AI progress. Great examples are RL from human feedback, which was key to training early AI assistants, and the aforementioned chain-of-thought monitoring, which was enabled by advances on reasoning models. We must focus the increasingly automated research process on developing new such insights, algorithms and theories, and iteratively build up safety cases for more capable AIs.

Scaling AI systems has to be constrained by our confidence in safety. We need to evolve commitments like the Preparedness Framework or Responsible Scaling Policy into widely mandated safety bars for continued development. These can be enforced by a network of third-party auditors, by government agencies or by international bodies.

The core challenge of automating AI research is not "getting there" - it is getting there in a way that keeps people a part of the continued improvement process, and leaves the future in humanity's hands.

What is next?

As we outlined recently with Sam, OpenAI prioritizes work in service of three north stars:

1. Navigating the next period of AI progress, by building an automated AI researcher, iterating with it on the alignment problem and finding ways for people to remain part of the self-improvement loop.

2. Delivering the benefits of scientific progress and economic growth that very intelligent machines enable.

3. Empowering everyone individually with a personal AGI.

user loses their admin status in the middle of us handling one of their admin requests?", or "what if the cache in storage partition A is updated a nanosecond before the one in storage partition B, leading some users to see different data?". Particularly if run in a loop or told to "make no mistakes", they will pile on cases like these until they've transformed your four-line change into a thousand-line monster.

This is why I'm confused by some people's claims that code review is dead. About half the time I review a large AI-generated change, I notice some unnecessary requirement that makes the diff more complex. When I dig deeper, the LLM invariably introduced this requirement on its own.

When working with a cache, I will often spell out in my prompt that I don't mind the data sometimes being stale or briefly inconsistent between cache instances. When reviewing Golang code, the first thing I do is scan for added mutexes or layers of indirection that don't look necessary to me.

If you allow requirements like these to propagate unchecked through a codebase, they will petrify the system over time. The whole thing will be such a mess of checking and re-checking and holding locks and enforcing consistency that it cannot be scaled at all. Keeping systems elegant requires a ruthless focus: on accepting stale data or occasional tail latency, on avoiding all but critical instances of coordination, and in general on figuring out what requirements you *don't* have to satisfy.

Control the ideas, not the code

ANTIREZ.COM · 18 JUL 2026 · SOURCE

```
Look at the past history of this blog. There are many blog posts abou

So mine is a trick. People feel more and more programming is complete

This is why yesterday, on X, I said that I believe many programmers a

1. You can now generate a lot of code, even *not* accounting for the

2. LLMs are very good at writing locally optimal code, and are worse

3. The working day is 8 hours. If you read the code, it is a tradeoff
```

Why dropping requirements is hard

[SVG OMITTED]

Anyone who's spent time designing real-world software (or who's studied distributed systems) knows that system design is a process of compromise. If you attempt to satisfy every single potential requirement — speed, concurrency, up-to-date data, a simple interface, supporting complex user behaviors — you will fail to design a working system at all.

Figuring out what requirements you can relax is difficult because **it combines both business and technical expertise**. It is technically challenging to identify precisely the level of staleness that a particular design change will allow; it is likewise challenging from a business perspective to determine whether that level of staleness is acceptable to users.

In most cases, an engineer who understands the technical side must try and explain it to a product manager, who can make a judgement on the business side. But this is a very difficult partnership to get right. Engineers and PMs have different incentives and little ability to check each other's work, which makes it hard to build trust. Without trust, the product manager is liable to avoid giving any ground at all, which leaves the design in an impossible state.

This is an underappreciated reason why solo developers or startups can sometimes run rings around larger companies. Because the technical and business decisions are united in a single person, they can make design tradeoffs that would be impossible for a large company, enabling them to build systems that are far more elegant and powerful.

Working with LLMs

[SVG OMITTED]

Knowing what requirements you don't have becomes far more important when working with coding agents. Frontier models are very capable, but they have all gone through the purifying flames of RLVR: like high-achieving schoolchildren, they are obsessed with satisfying the Grader. On top of that, they have very little business context compared to an experienced human. That often manifests as them catching correctness or consistency "bugs" that are really just hypothetical unsatisfied requirements.

For instance, it's not uncommon for an LLM to casually introduce a mutex or other concurrency mechanism to ensure that two pieces of semi-related data are kept in sync, even if not told to do so. I've seen LLMs add weird checks in the middle of request handling to ensure that some already-checked piece of data hasn't changed mid-flight, even if that wouldn't successfully prevent the feared TOCTOU bug. LLMs are obsessed with edge cases like "what if the current

I have focused in this essay only on the first point, as I believe it is by far the most urgent. However, I hold a deep hope and appreciation for the benefits that further technological progress will bring. Future aligned AI could advance science, develop new therapies, and bring about broad material abundance. Friendly and honest AI can help people navigate difficulties they face in their life and meaningfully improve their happiness and sense of fulfillment. OpenAI puts a tremendous amount of effort into bringing these benefits about. One current example I am proud of - and my loved ones have found helpful - is the deep investment into ChatGPT's ability to provide health information.

As great as the long-term promise of AI may be, the majority of our focus should be on the next few years. We are facing a transition to a world with incredibly intelligent machines, and we need to ensure that transition works out well for humanity. We need to find ways to preserve human agency and enshrine an intrinsic value to being human, in a world where most tasks could be performed by AI. To prevent extreme concentration of power in a world where undertakings that would have taken thousands of experts now will be achievable by a few people operating a large computer. And to ensure that humans remain in control of the future and are not left behind by unchecked progress, brought about by an alien intellect exceeding our own.

Currently I believe that no lab has solved alignment and monitoring to a sufficient degree to continue responsibly scaling at maximum speed for much longer. I expect and hope for voluntary slowdowns to become commonplace until shared safety bars are established. And I believe that international coordination on future AI development needs to become a top priority for governments around the world.

- Alignment
- 2026

Author

Jakub Pachocki

Footnotes

1. This included scaling self-play(opens in a new window), robotics(opens in a new window), and, in retrospect most notably, scaling recurrent networks to model language(opens in a new window), which was a precursor to the GPT line of research.

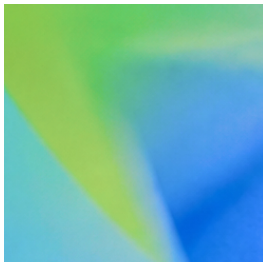
   [find in text]

2. A secondary reason for this design was preventing distillation. However, maintaining CoT monitorability has explicitly been the bigger priority for us throughout development.

   [find in text]

Keep reading

View all



Research acceleration: The view inside OpenAI

ResearchSep 6, 2026

[VIDEO OMITTED]

GPT-6 Astra: A new generation of intelligence

ResearchSep 3, 2026



Safety overview: GPT-6 Astra

SafetySep 3, 2026

What requirements don't you have?

SEANGOEDECKE.COM RSS FEED · 19 OCT 2026 · SOURCE

When designing software, the requirements you don’t have are as important as the requirements you do. Identifying what you can compromise on — what can be slow, or stale, or inflexible — is usually the key to an elegant design.

Elegant systems drop some requirements

[SVG OMITTED]

For instance, take preforking web servers like Unicorn. Because they lack the feature of shared mutable state between server instances, they can run each server in its own process, which provides isolation for free. Because Unicorn specifically ignores the problem of slow client handling, it can queue requests directly on Unix sockets, drastically simplifying the implementation.

Because S3 does not support atomic multi-key updates — you cannot update multiple blobs in a single transaction — it can store data at the blob level however it chooses. That makes it much faster, since reads and writes don’t have to cross storage partitions, and much easier to scale, since less coordination means fewer sources of truth.

StatsD uses UDP by default, which means it doesn’t guarantee that every piece of data you send makes it into storage. In exchange for dropping that requirement, it incurs much less network overhead, which is a big advantage for a stats aggregation service that’s constantly shoveling data over the wire.

The general design approach of fanning out data to an edge cache buys a lot of speed and scaling by dropping the requirement that data must always be up-to-date. SQLite buys operational simplicity — it can run out of a single file! — by dropping the requirement of concurrent writes. By accepting high tail latency, Cursor can scale their distributed git backend more easily. Well-designed systems turn their weaknesses into strengths.